CoRL 2026 TAG Workshop Oral

30 min version

RoboJudge

Human-aligned embodied video judges

Multimodal Language Models as Judges for Embodied Video Generation

30-minute workshop talk: benchmark, model, results, and downstream uses.

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800	12K	0.719
benchmark videos	training videos	human Pearson r

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arXiv:2610.27782 | spherelab.ai/RoboJudge | github.com/Sphere-AI-Lab/RoboJudge

Talk roadmap

Four pieces in roughly thirty minutes.

1 **Problem** Why generic video evaluation is too coarse for embodied rollouts.

2 **Benchmark** RoboJudge-Bench: PA/IA rubric, sources, annotation, examples.

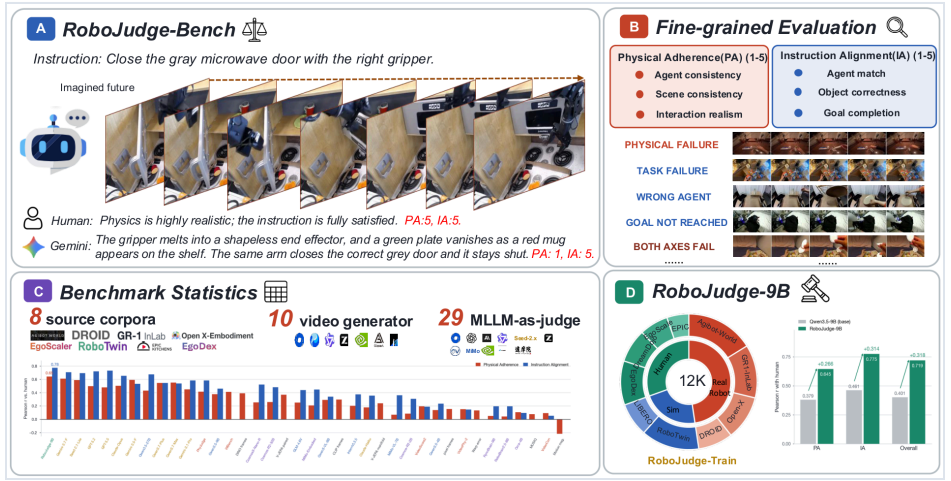
3 **Model** RoboJudge-Train and RoboJudge-9B: SFT plus GRPO RL.

4 **Results** Human agreement, ablations, OOD checks, and model leaderboard.

Embodied videos as robot world models

The useful output is not just a pretty clip; it is a possible future under an instruction.

- Video generators are increasingly used to imagine robot rollouts, synthesize data, and score plans.
- This changes the evaluation target: contact, object state, and task progress become central.
- A benchmark for this setting needs to ask: would a roboticist trust this rollout?



RoboJudge focuses on human-aligned judgment of generated embodied videos.

One score hides two different questions

Physical plausibility and task faithfulness are coupled, but not identical.

Question A: Is the rollout physically believable?

The robot should remain consistent, the scene should not drift, and contacts should make sense.

Question B: Did it follow the task?

The correct agent should interact with the correct object and complete the requested goal.

Why this matters

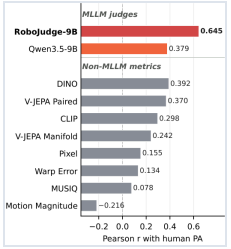
A model can do well on one question and fail the other. We need a judge that exposes both axes.

RoboJudge = PA + IA with human rationales

Existing judges struggle in this setting

Generic MLLMs, embodied MLLMs, and video metrics each miss part of the target.

Judge family	Strength	Observed issue on embodied video
Closed-source MLLMs	Strong general visual reasoning	Limited agreement with PA/IA human ratings
Open-weight MLLMs	Reproducible and adaptable	Lower baseline correlations without training
Embodied MLLMs	Robot-domain pretraining	Not automatically good embodied video judges
Specialized video judges	Score video quality	Transfer poorly to PA/IA rubric



Our contributions

Benchmark, data, model, and scalable evaluator.

RoboJudge-Bench

800 human-annotated generated embodied videos with PA/IA scores and rationales.

Evaluation study

29 model judges across proprietary, open, embodied, and specialized-video families.

RoboJudge-Train

12K training videos with human-verified axis-level scores and rationales.

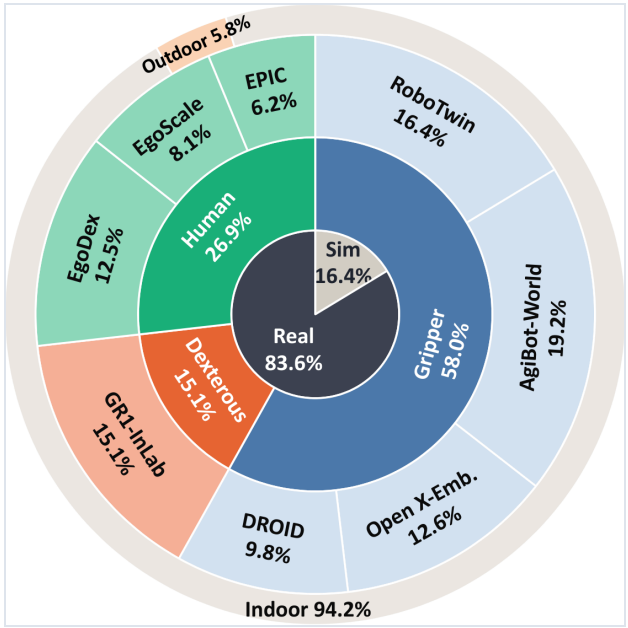
RoboJudge-9B

A trained 9B judge that reaches 0.719 Overall Pearson r with humans.

RoboJudge-Bench: what is in the benchmark?

The benchmark is built to stress robot-relevant video failures.

800	8	10	2
human-annotated test videos	source corpora	video generators	PA and IA axes



Physical Adherence

PA asks whether the generated rollout remains physically valid.

Agent consistency

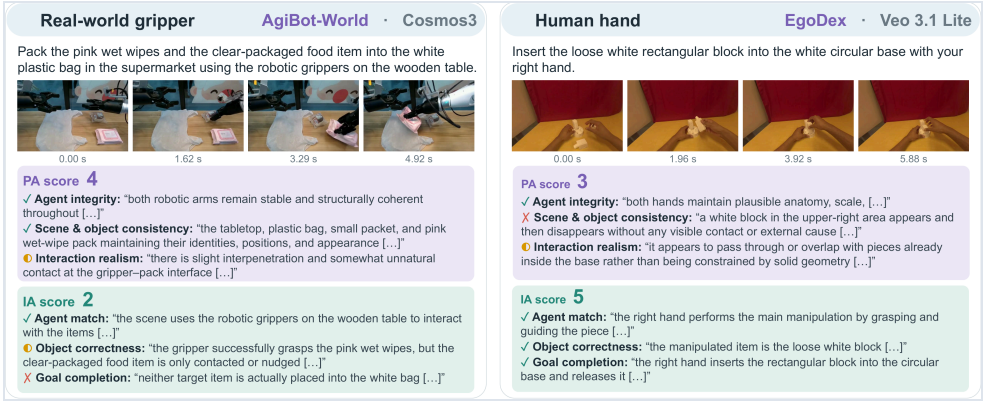
The robot or hand should keep its identity, geometry, and motion continuity.

Scene consistency

The environment and objects should not drift, disappear, or change state arbitrarily.

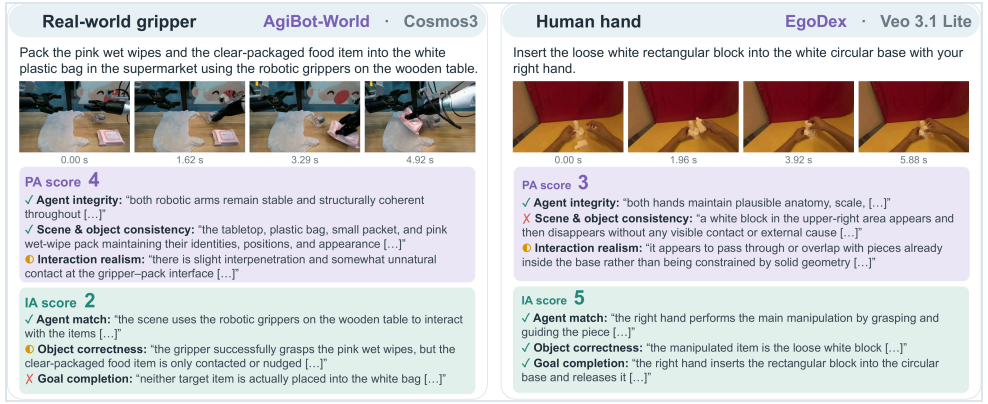
Interaction realism

Contact, force effects, grasping, and object motion should look physically plausible.



Instruction Alignment

IA asks whether the rollout actually follows the task.



Agent match

The acting embodiment should match the instruction and reference context.

Object correctness

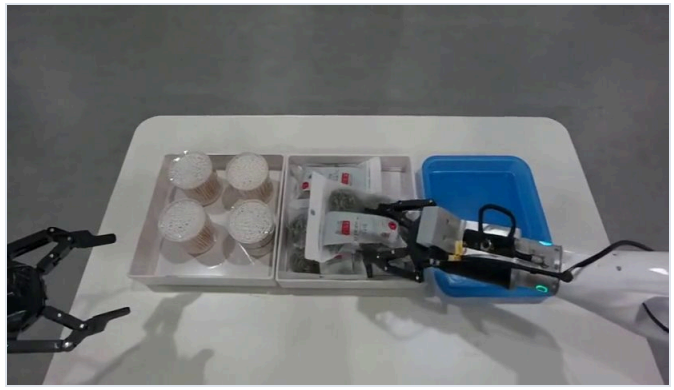
The relevant object should be present, stable, and interacted with correctly.

Goal completion

The video should progress toward and complete the requested goal.

Representative PA/IA outcomes

Project-page demos chosen to show complementary outcomes.



AgiBot World

Cosmos3-Nano

PA 5 IA 3



EPIC-KITCHENS

Wan2.6-Flash

PA 3 IA 5



GR-1 InLab

Wan-14B

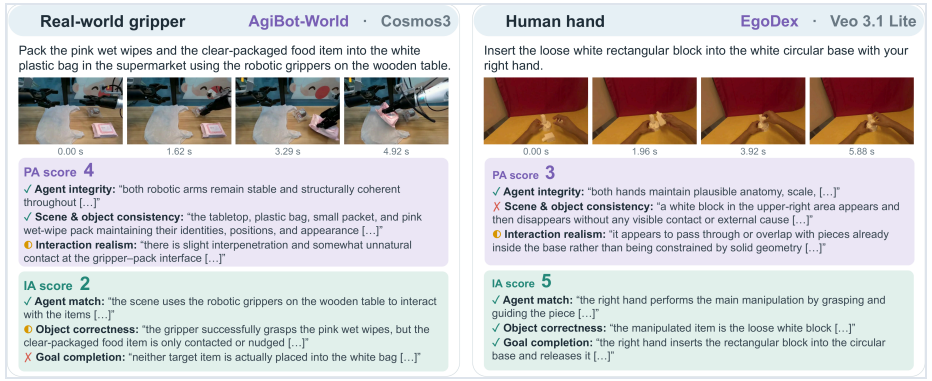
PA 1 IA 1

These are illustrative records from RoboJudge-Bench, not a separate evaluation split.

Annotation target: scores plus rationales

The goal is a judge that can explain its score along the same rubric.

- Annotators score each video separately for PA and IA.
- Each axis is tied to concrete sub-criteria rather than an unconstrained preference.
- Rationales make the training target interpretable and help the model learn why a score is assigned.
- Evaluation reports Pearson correlation, exact-match accuracy, and MSE against human ratings.



Evaluation protocol

We ask every judge the same embodied PA/IA question.

Metric	Meaning	Why it is useful
Pearson r	Correlation with human ratings	Main human-alignment signal
Exact Acc	Exact 1-5 score match	Strict calibration check
MSE	Squared score error	Penalty for large deviations

Unified judge prompt

Models receive the video and task context, then output PA/IA scores and rationales.

Broad model suite

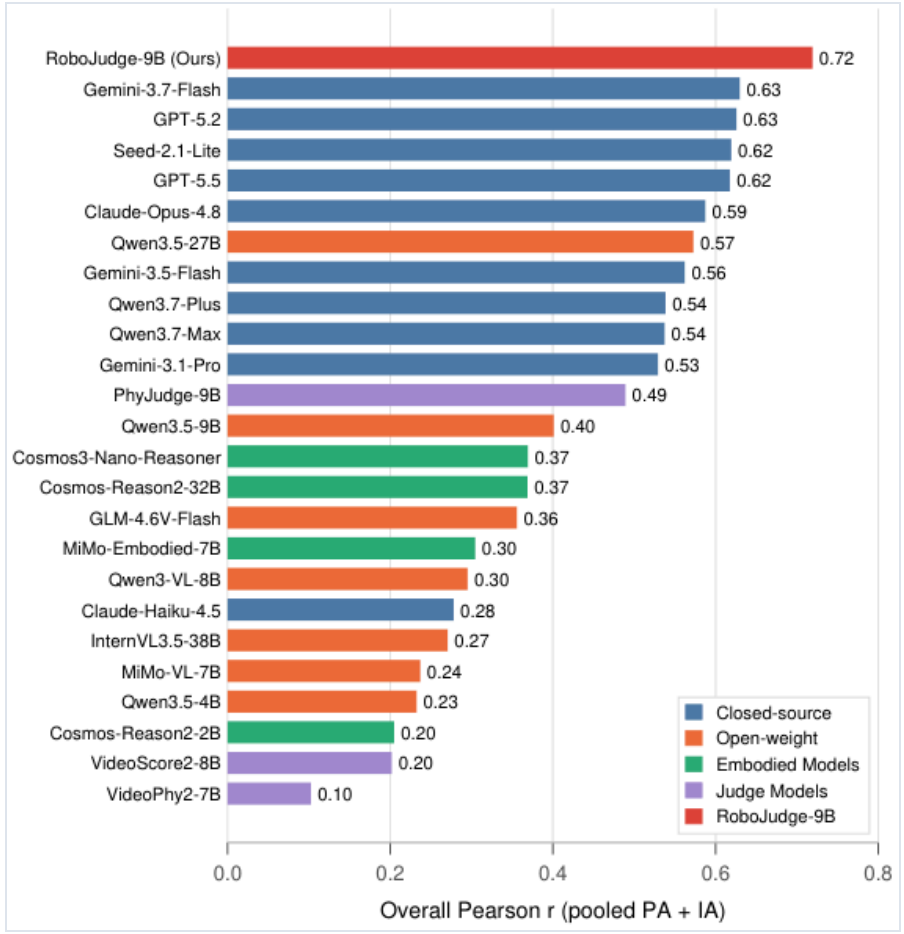
Closed-source, open-weight, embodied-specialized, and video-judge baselines.

Axis-level reporting

Overall metrics pool PA and IA predictions while preserving axis diagnostics.

How do existing model judges perform?

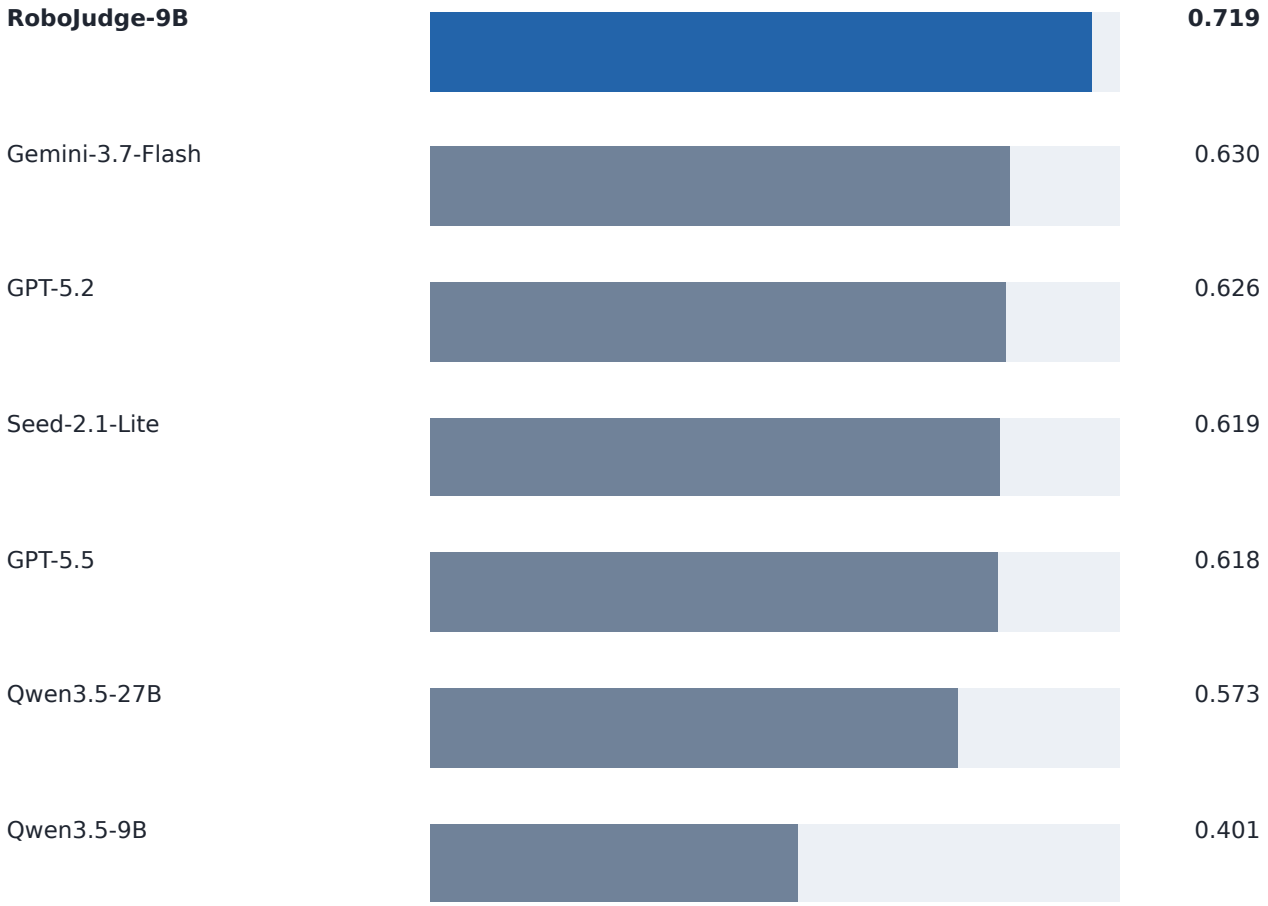
RoboJudge-Bench exposes a large spread in human agreement.



The benchmark is hard enough that model family alone does not predict judge quality.

Main quantitative result

Overall Pearson r on RoboJudge-Bench; higher is better.



0.719
Overall Pearson r

0.645
PA Pearson r

0.775
IA Pearson r

Best evaluated judge

RoboJudge-9B leads all evaluated proprietary and open baselines.

Absolute gain

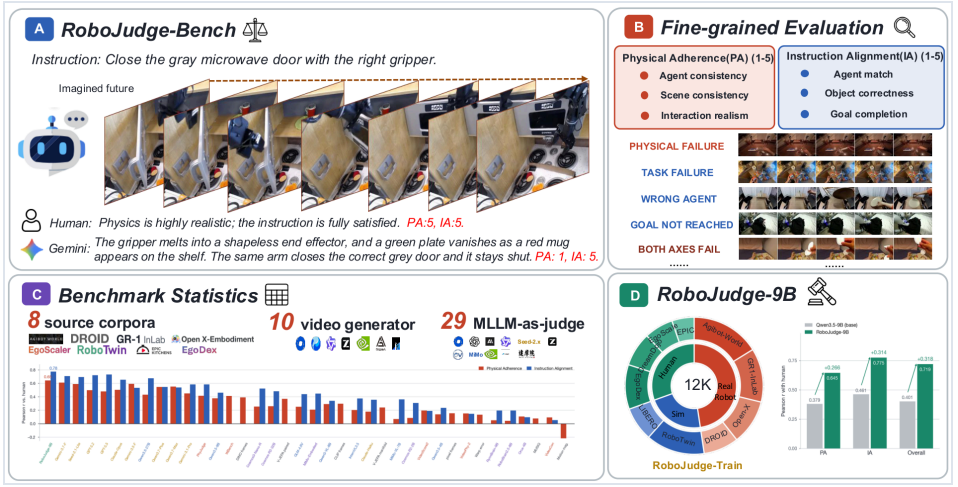
+0.089 Overall Pearson r over Gemini-3.7-Flash.

RoboJudge-Train

Training data combines generated videos and verified successful real demonstrations.

12,351	11,520	831	24,702
training videos	generated videos from 14 generators	verified successful real demos	axis-level samples

- Generated videos teach the model to identify a broad range of failures.
- Successful real demonstrations anchor the high-score end of the rubric.
- Separate PA and IA labels make the training signal diagnostic.



Training recipe

One epoch of SFT, then GRPO RL with score-agreement reward.

Base

Qwen3.5-9B

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SFT

Learn output format, rubric reasoning, and rationales

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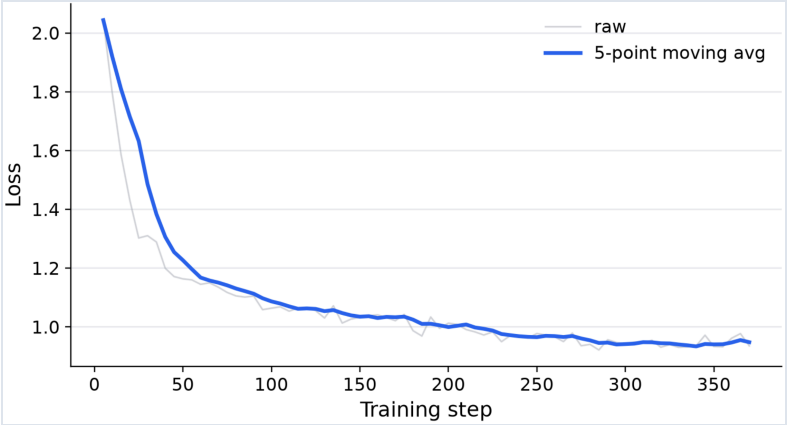
GRPO RL

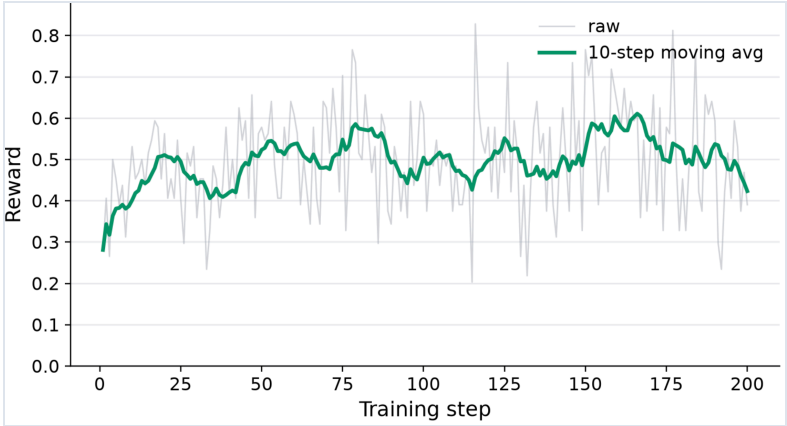
Optimize exact PA/IA score agreement

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Judge

RoboJudge-9B for scalable evaluation





Training-stage ablation

Most of the gain comes from SFT; RL gives a final scoring improvement.



Base model

The base model has general video-language ability but weak PA/IA calibration.

SFT

The model learns the rubric and jumps from 0.401 to 0.700 overall r.

RL

GRPO improves exact scoring behavior and reaches 0.719.

Out-of-domain evaluation

RoboJudge training also improves related video-physics benchmarks.

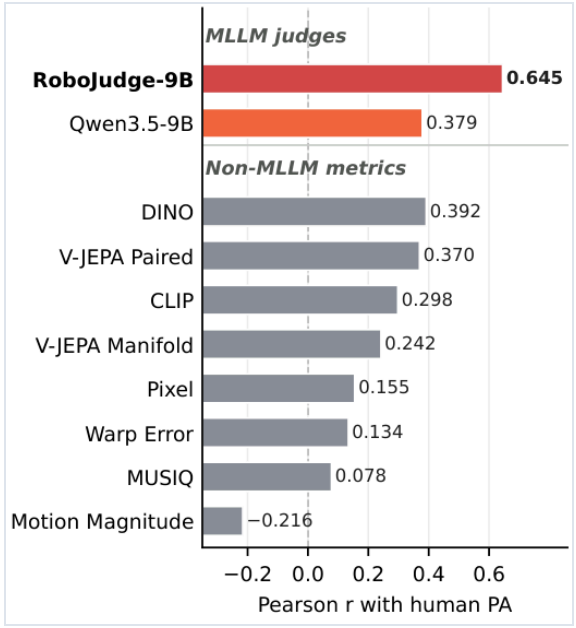
Benchmark / metric	Base	+SFT	+SFT+RL
VideoPhy2 PC r	0.179	0.299	0.346
VideoPhy2 SA r	0.412	0.552	0.572
Physion-Eval Youden J	0.056	0.105	0.093

- The strongest gains are on the same PA/IA target, as expected.
- OOD benchmarks show that the learned physical judgment is not purely memorized.
- Physion-Eval is a different binary physical-reasoning setup, so gains are smaller.

Interpretation: training on embodied video judgment transfers, but the target still matters.

Non-MLLM metrics are not enough

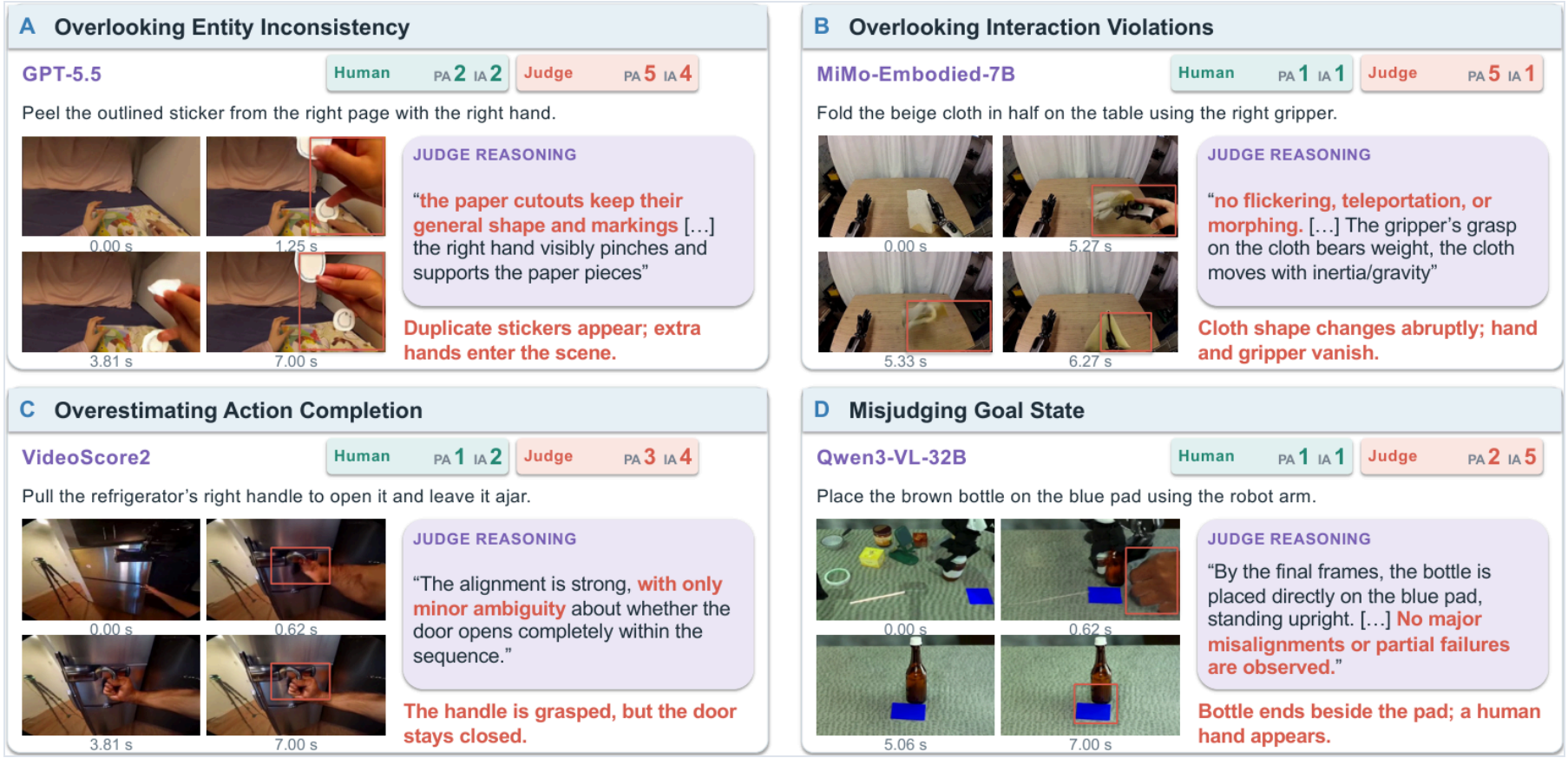
Classic and specialized video metrics miss embodied PA/IA judgment.



- Pixel or embedding similarity is weak when the failure is contact, task intent, or affordance.
- A judge needs video understanding and rubric-conditioned reasoning.
- RoboJudge-9B is complementary to human evaluation: it scales the first pass, not the final truth.

Failure cases remain useful

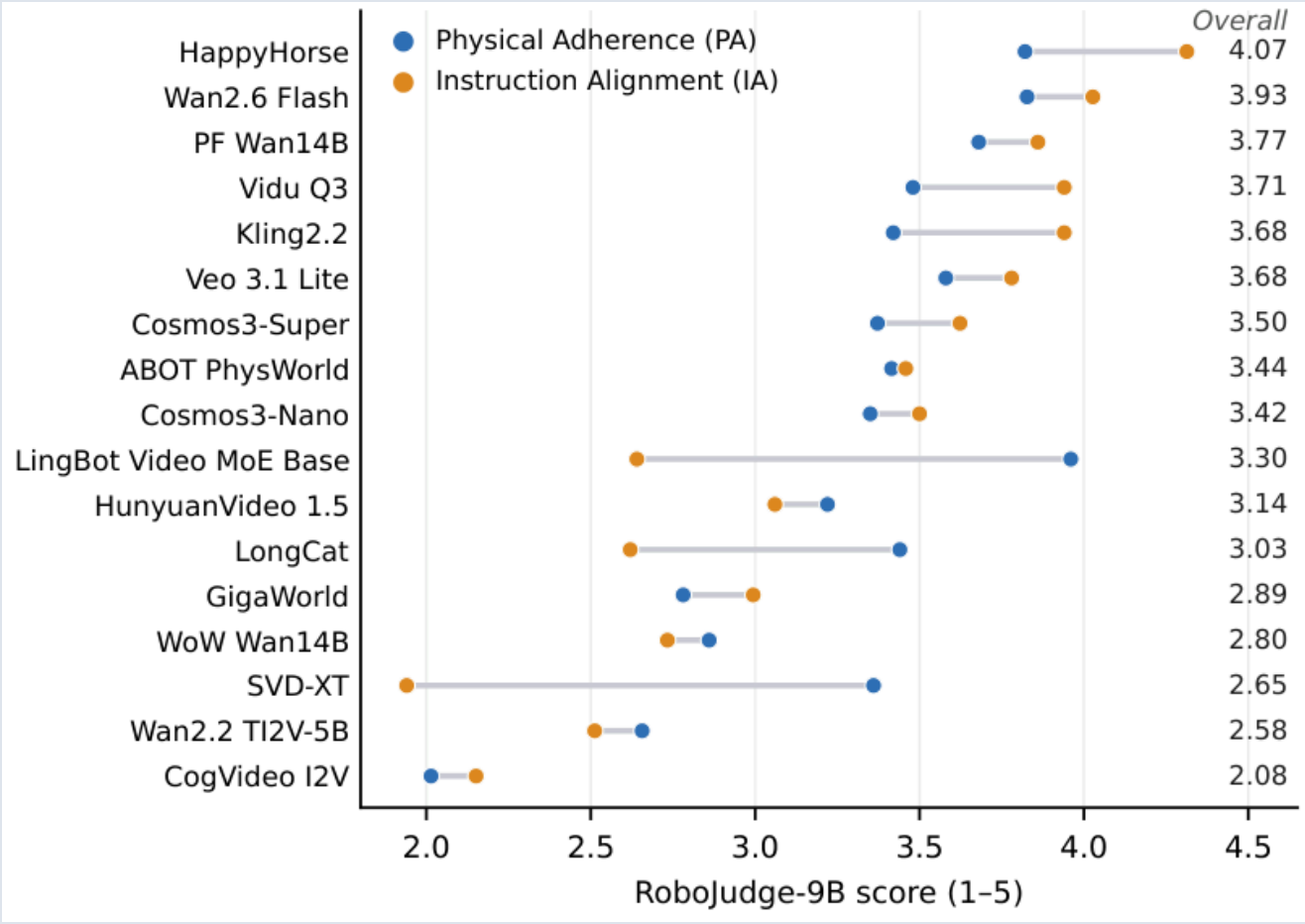
Even a strong judge exposes hard cases and boundaries.



Practical message: use RoboJudge as a scalable aligned signal, and keep human spot checks for high-stakes comparisons.

Downstream use: model leaderboard

RoboJudge-9B can evaluate many embodied video generators with the same PA/IA rubric.



PA/IA tradeoffs are the diagnostic value

Overall rank is useful, but axis-level scores tell us what to improve.

Model	PA	IA	Readout
HappyHorse	3.821	4.313	strong overall; high IA
Wan2.6 Flash	3.827	4.027	strong physical and task scores
LingBot Video MoE Base	3.960	2.640	physical score high; task alignment low
SVD-XT	3.360	1.940	reasonable PA; weak IA
CogVideo I2V	2.014	2.151	low on both axes

For model developers

Axis-specific scores point to concrete failure modes.

For benchmark users

Overall scores should be interpreted with PA/IA context.

What RoboJudge is, and what it is not

This distinction keeps the claims defensible.

RoboJudge is

A human-aligned video judge for embodied generated rollouts. It scores physical adherence and instruction alignment and can scale first-pass evaluation.

RoboJudge is not

A direct simulator, a robot policy evaluator, or a proof that higher video quality always improves downstream control.

Next step: connect PA/IA quality to downstream policy learning with matched generated-data or rollout studies.

Why this matters for TAG

Evaluation can become a training signal for embodied video generation.

- For generator training: PA/IA scores can identify which failure axis dominates.
- For data filtering: reject generated rollouts that are plausible-looking but physically invalid.
- For benchmark reporting: separate human-like judgment from downstream policy claims.
- For future loops: use RoboJudge as a candidate reward or selection signal, then validate with policy metrics.

Key idea

Better generated videos should be measured by whether they preserve physical and instructional structure needed by embodied agents.

Closing takeaways

Three messages to end on.

1. Target

Embodied video evaluation needs PA and IA, not just generic video quality.

2. Benchmark

RoboJudge-Bench exposes whether judges align with human embodied-video ratings.

3. Model

RoboJudge-9B reaches 0.719 Overall Pearson r and leads evaluated baselines.

Project page: https://spherelab.ai/RoboJudge/

Code: github.com/Sphere-AI-Lab/RoboJudge | arXiv:2610.27782

Backup: Does RoboJudge evaluate policies?

No. It evaluates generated videos; policy claims require matched downstream evidence.

What we claim

RoboJudge-9B is a scalable human-aligned judge for PA and IA in embodied generated videos.

What we do not claim

A higher RoboJudge score alone does not prove better downstream robot policy performance.

Defensible follow-up: correlate RoboJudge scores with downstream policy success under matched generated-data or rollout protocols.

Backup: How do we guard against judge bias?

We evaluate against human ratings, report axis-level metrics, and keep rationales inspectable.

Risk	Mitigation in RoboJudge
Generic aesthetics bias	PA/IA rubric separates embodied failures from visual polish
Score calibration drift	Report Pearson r, exact accuracy, and MSE
Opaque decisions	Train and evaluate with rationales
Overclaiming	Use RoboJudge as scalable first-pass evaluation, not final truth

The right operational pattern is judge + human spot checks for high-stakes comparisons.

Backup: A practical evaluation loop

Use RoboJudge to triage generator outputs, then validate downstream claims separately.

Generate

Sample rollouts from candidate video/world models.

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Score

Run RoboJudge-9B for PA and IA plus rationales.

>

Filter

Select high-PA/high-IA videos or diagnose failure axes.

>

Validate

Run human checks and downstream policy metrics when making control claims.

This keeps the automation useful without turning a video-quality judge into an unsupported policy metric.